

Processando config 14, seed 3, teste: Alphabit
xx
Starting Alphabit: nb = 16777216
Version: TestU01 1.2.3
xx

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed3.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 2,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L = 4$
Expected number per cell = 4194304
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.9802323e-08, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom	:	2
Value of the statistic	:	2.30
p-value of test	:	0.32

CPU time used : 00:00:00.01

Generator state:
16777216 bits have been read.

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 4,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = $2^L = 16$
Expected number per cell = 1048576
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 2.0861626e-07, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 8
Value of the statistic : 6.74
p-value of test : 0.56

CPU time used : 00:00:00.00

Generator state:
33554432 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 8,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 256
Expected number per cell = 65536
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 3.784895e-06, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 128
Value of the statistic : 130.89
p-value of test : 0.41

CPU time used : 00:00:00.01

Generator state:
50331648 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed3.bin

smultin_MultinomialBitsOver test:

N = 1, n = 16777216, r = 0, s = 32, L = 16,
Sparse = FALSE

Number of bits = n = 16777216
Number of cells = 2^L = 65536
Expected number per cell = 256
Hashing = FALSE

For Delta > -1, we use the ChiSquare approximation
Correction factor of the ChiSquare:

Delta = 1, Mu = 0.00097653271, Sigma = 1

Test Results for Delta = 1.0000

Number of degrees of freedom : 32768
Value of the statistic : 32634.98
p-value of test : 0.70

CPU time used : 00:00:00.01

Generator state:

67108864 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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sstring_HammingIndep test:

N = 1, n = 524288, r = 0, s = 32, L = 16, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 177
Chi-square statistic : 147.70
p-value of test : 0.95

CPU time used : 00:00:00.01

Generator state:

83886080 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de

Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed3.bin

sstring_HammingIndep test:

N = 1, n = 262144, r = 0, s = 32, L = 32, d = 0

Counters with expected numbers >= 10

Number of degrees of freedom : 305
Chi-square statistic : 279.44
p-value of test : 0.85

CPU time used : 00:00:00.00

Generator state:

100663296 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed3.bin

sstring_HammingCorr test:

N = 1, n = 524288, r = 0, s = 32, L = 32

Normal statistic : 0.46
p-value of test : 0.32

CPU time used : 00:00:00.00

Generator state:

117440512 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

ufile_CreateReadBin: /home/matheus/Área de
Trabalho/qanalog_prng-main/qanalog_prng-main/outputs/out_config14_seed3.bin

swalk_RandomWalk1 test:

N = 1, n = 262144, r = 0, s = 32, L0 = 64, L1 = 64

Test on the values of the Statistic H

Number of degrees of freedom : 32
ChiSquare statistic : 29.00
p-value of test : 0.62

Test on the values of the Statistic M

Number of degrees of freedom : 31
ChiSquare statistic : 32.29
p-value of test : 0.40

Test on the values of the Statistic J

Number of degrees of freedom : 32
ChiSquare statistic : 39.07
p-value of test : 0.18

Test on the values of the Statistic R

Number of degrees of freedom : 25
ChiSquare statistic : 25.91
p-value of test : 0.41

Test on the values of the Statistic C

Number of degrees of freedom : 15
ChiSquare statistic : 4.25
p-value of test : 0.9968

CPU time used : 00:00:00.07

Generator state:
134217728 bits have been read.

HOST = matheus-IdeaPad-3-15ALC6, Linux

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swalk_RandomWalk1 test:

N = 1, n = 52428, r = 0, s = 32, L0 = 320, L1 = 320

Test on the values of the Statistic H

Number of degrees of freedom : 62

ChiSquare statistic : 38.25
p-value of test : 0.9924

Test on the values of the Statistic M

Number of degrees of freedom : 62
ChiSquare statistic : 90.72
p-value of test : 0.01

Test on the values of the Statistic J

Number of degrees of freedom : 160
ChiSquare statistic : 186.04
p-value of test : 0.08

Test on the values of the Statistic R

Number of degrees of freedom : 57
ChiSquare statistic : 51.37
p-value of test : 0.69

Test on the values of the Statistic C

Number of degrees of freedom : 32
ChiSquare statistic : 24.72
p-value of test : 0.82

CPU time used : 00:00:00.03

Generator state:
150994688 bits have been read.

===== Summary results of Alphabit =====

Version: TestU01 1.2.3
Generator: ufile_CreateReadBin
Number of bits: 16777216
Number of statistics: 17
Total CPU time: 00:00:00.22

All tests were passed